

Tsung-Shan (Kevin) Yang

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Education

University of Southern California (USC) Ph.D. in Electrical Engineering	Aug 2022 – May 2026
• Advisor: Prof. C-C. Jay Kuo • Thesis: Interpretable and Efficient Multi-Modal Data Interplay: Algorithms and Applications	
National Taiwan University (NTU) B.S., M.S. in Electrical Engineering	Sep 2014 – Jun 2021
National Taiwan University (NTU) B.S. in Chemistry	Sep 2014 – Jun 2019

Experience

Machine Learning Engineer , LinkedIn Corp. – Sunnyvale, CA	Jun 2026 – Present
• Evaluate the rubric generation model for the LinkedIn search agent	
Machine Learning Engineer , Tiktok Inc. – San Jose, CA	May 2025 – May 2026
• Developed a SOTA AI-generated video detection with 3% of parameters and a 98% reduction in inference time • Accelerated hack-detection pipeline 10× by identifying malicious users and abnormal video patterns	

Selected Publications

[C1] SVD-Det: A Lightweight Framework for Video Forgery Detection Using Semantic and Visual Defect Cues Tsung-Shan Yang , Tianyu Zhang, Feng Qian, Bing Yan, C.-C. Jay Kuo <i>Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) (2026)</i> • 97% smaller, 98% faster, while achieving +2.7% higher AUC for AIGC video forgery detection
[J1] Efficient Human-Object-Interaction Detection via Interaction Label Coding and Conditional Decision Tsung-Shan Yang , Yun-Cheng Wang, Chengwei Wei, Suyu You, C.-C. Jay Kuo <i>Computer Vision and Image Understanding (CVIU) (2025): 104390.</i> • Reduce computational cost by 15,800× fewer FLOPs compared to state-of-the-art methods
[C2] Challenges in video-text retrieval: via transparent alignment Tsung-Shan Yang , Yun-Cheng Wang, Chengwei Wei, Suyu You, C.-C. Jay Kuo <i>Applications of Digital Image Processing XLVIII. Vol. 13605. SPIE, 2025</i> • Deploy an explainable ranking pipeline for video-text retrieval, reducing 97% trainable parameters
[J2] Image-Text Retrieval via Green Explainable Multi-modal Alignment (GEMMA) Tsung-Shan Yang , Yun-Cheng Wang, Chengwei Wei, Suyu You, C.-C. Jay Kuo <i>APSIPA Transactions on Signal and Information Processing (2025)</i> • Developed an interpretable alignment framework for image and text encoders with 3% of trainable parameters
[C3] BPQA: A Blind Point Cloud Quality Assessment Method Qingyang Zhou, Aolin Feng, Tsung-Shan Yang , Shan Liu, C.-C. Jay Kuo <i>IEEE International Conference on Image Processing Challenges and Workshops (ICIPCW), 2023</i> • Develop an interpretable learning framework with minimal computational overhead
[J3] Viewing Bias Matters in 360 Videos Visual Saliency Prediction Peng-Wen Chen, Tsung-Shan Yang , Gi-Luen Huang, Chia-Wen Huang, Yu-Chieh Chao, Pei-Yuan Wu <i>IEEE Access Journal paper, 2023</i> • Analyzed human bias in saliency maps and extended spherical kernels to time-series data
[C4] NTIRE 2020 Challenge on NonHomogeneous Dehazing <i>Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops. (CVPRW) 2020.</i> • Propose an attention refinement block of the deep learning model

Awards/Scholarships

2024 IEEE MIPR Student Grant	<i>IEEE TCMC</i>
2022 Taiwan - USC Scholarship	<i>Ministry of Education in Taiwan</i>
2022 Viterbi School of Engineering / Graduate School Fellowship	<i>University of Southern California</i>
2014 Fall & 2015 Spring Dean's List	<i>National Taiwan University</i>
2011 Gold Medal	<i>International Junior Science Olympiad</i>

Teaching Experience

Systems for Machine Learning , University of Southern California	2024 Spring, 2025 Spring
• Introduce the hardware of TPUs and GPUs • Design the project about LLM inference, such as LoRA and KV-cache	
Introduction for Programming , University of Southern California	2024 Fall
• Lead weekly hand-on labs • Introduce good coding styles and algorithms	
Machine Learning , National Taiwan University	2019 Fall, 2020 Fall
• Design assignments about theoretical analysis and deep learning projects • Maintain the course website	
Data Structure , National Taiwan University	2020 Spring
• Design assignments about theoretical analysis and data structure implementation	
General Chemistry , National Taiwan University	2018 Fall
• Lead group discussions and provide hints on assignments • Provide two-hour TA classes each week for over 300 students	

Technologies & Expertise

Core Expertise: Multimodal Alignment, Video Understanding, Interpretable and Efficient Deep Learning

Research Strengths: Lightweight and explainable model design, Vision-language retrieval and ranking systems

Programming Languages: Python, C++, C, MATLAB, HTML

Frameworks & Libraries: PyTorch, OpenCV, TensorFlow, Keras, Scikit-learn

Languages: English (Professional Proficiency), Mandarin (Native)

Code & Projects: Research projects and implementations available at: <https://github.com/keevin60907>